Year 13 Further Maths - Series

Similar questions - Method of differences

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 In this question you must show detailed reasoning.

(a) Use partial fractions to show that $\sum_{r=1}^{n} \frac{1}{r^2+5r+6} = \frac{1}{3} - \frac{1}{n+3}$. **[5]**

(b) Hence find $\sum_{r=1}^{\infty} \frac{1}{r^2+5r+6}$. **[1]**

2 Using an algebraic method, determine the least value of n for which $\sum_{r=1}^{n} \frac{1}{(2r-1)(2r+1)} \geqslant 0.495$. **[8]**

Question 1

(a) $\dfrac{1}{(r+2)(r+3)} = \dfrac{1}{r+2} - \dfrac{1}{r+3}$. Sum $= \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \ldots + \left(\frac{1}{n+2} - \frac{1}{n+3}\right) = \frac{1}{3} - \frac{1}{n+3}$

(b) As $n \to \infty$, $\frac{1}{n+3} \to 0$, so the sum is $\frac{1}{3}$

Question 2

$\dfrac{1}{(2r-1)(2r+1)} = \dfrac{1}{2}\left(\dfrac{1}{2r-1} - \dfrac{1}{2r+1}\right)$. The sum is $\dfrac{1}{2}\left(1 - \dfrac{1}{2n+1}\right) = \dfrac{n}{2n+1}$. Then $\dfrac{n}{2n+1} \geqslant 0.495$ gives $n \geqslant 0.99n + 0.495$, so $0.01n \geqslant 0.495$ and $n \geqslant 49.5$. The least value is $n = 50$ (check: $\frac{50}{101} = 0.49505$, $\frac{49}{99} = 0.49495$)